Code No: 135AF **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, August - 2022
DESIGN AND ANALYSIS OF ALGORITHMS
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Explain about the types of Randomized algorithm.
b) Write an algorithm of Quick sort and analyze performance of quick sort in worst case. [7+8]

2.a) What are asymptotic notations? Explain with examples.
b) Sort the following elements using Merge sort: [7+8]
12, 2, 16, 30, 8, 28, 4, 10, 20, 6, 18

3.a) Explain the Union algorithm with weighing rule.
b) Discuss about the general method of backtracking. [7+8]

4.a) Write an algorithm to determine Bi-connected components.
b) Describe sum of subsets problem with an example. [7+8]

5.a) Show how Prim's algorithm can be implemented using heap. What would be the time complexity of the algorithm?
b) Write the applications of Greedy method. [10+5]

6. Write the Greedy algorithm for sequencing unit time jobs with deadlines and profits. Trace the algorithm with suitable example. [15]

7.a) Use the function OBST to compute w(i, j), r(i, j), and c(i, j), $0 \le i < j \le 4$, for the identifier set (a1,a2,a3,a4) = (cout, float, if, while) with p(1) = 1/20, p(2) = 1/5, p(3) = 1/10, p(4) = 1/20, q(0) = 1/5, q(1) = 1/10, q(2) = 1/5, q(3) = 1/20, and q(4) = 1/20. Using the r(i, j)'s construct the optimal binary search tree.
b) Discuss about chained matrix multiplication in dynamic programming. [10+5]

8. Explain the following:
a) LC branch and bound
b) Cook's theorem. [8+7]

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